

MUSSALA, Bulgaria

WMO: 156150

Lat: 42.1793N

Lon: 23.5851E

Elev: 9603

StdP: 10.26

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-3.7	-21.6	2.4	7.7	-15.3	3.4	4.9	76.2	19.3	63.4	18.6	21.3	20	1.025

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	57.0	46.8	54.4	45.6	52.0	44.4	48.9	53.8	47.3	52.0	45.9	50.2	9.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
46.8	68.0	50.5	45.1	63.8	48.9	43.6	60.2	47.6	23.1	54.2	22.1	52.2	21.2	50.4	59.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 62.7	44.9	36.1	-12.0	61.8	5.0	2.9	-15.6	63.9	-18.5	65.6	-21.4	67.2	-25.0	69.3		
(5)			-12.5	53.2	5.0	2.8	-16.1	55.2	-19.0	56.8	-21.9	58.4	-25.5	60.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	14.6	14.8	17.8	24.0	32.0	38.9	43.3	44.1	37.4	31.9	24.4	17.7		
	DBStd	12.71	8.37	8.46	7.51	6.83	5.77	6.43	5.74	5.33	6.35	7.45	8.81	8.20		
	HDD50	7877	1097	985	998	781	559	336	219	195	378	561	768	1000		
	HDD65	13325	1562	1405	1463	1231	1024	783	673	648	827	1026	1218	1465		
	CDD50	26	0	0	0	0	0	3	10	12	1	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	15.8	19.8	20.2	20.0	16.9	14.0	12.0	10.8	9.7	11.9	14.9	19.2	20.1		
	PrecAvg	31.3	2.7	2.8	3.3	3.6	3.5	2.8	2.6	2.0	2.0	1.7	1.6	2.6		
	PrecMax	39.8	6.7	8.2	7.5	10.0	5.9	6.1	5.5	5.2	6.5	4.1	4.9	5.4		
	PrecMin	21.3	0.8	0.5	1.2	1.6	1.4	0.5	0.6	0.3	0.4	0.2	0.0	0.2		
	PrecStd	5.6	1.6	1.9	1.5	1.8	1.2	1.2	1.3	1.4	1.2	1.0	1.0	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	34.8	36.2	42.3	47.9	57.9	61.5	60.2	54.7	51.3	43.9	38.6		
		MCWB	25.2	27.0	29.9	34.0	38.9	44.6	48.0	48.9	42.9	40.2	33.0	28.2		
	2%	DB	30.5	30.3	32.2	38.6	44.9	53.8	57.3	56.9	51.1	47.1	40.6	32.8		
		MCWB	23.4	25.0	27.7	32.5	37.6	44.2	47.2	47.5	42.0	37.1	30.8	25.9		
	5%	DB	27.7	27.5	29.4	35.1	42.2	50.8	54.3	54.3	48.6	44.0	37.6	30.1		
		MCWB	22.4	23.4	26.0	30.7	36.9	43.2	46.0	46.0	40.6	35.7	29.4	25.4		
	10%	DB	25.0	24.9	27.3	32.2	39.8	48.0	51.7	51.8	45.9	41.3	34.7	27.9		
		MCWB	21.4	21.6	24.6	29.8	35.6	42.0	44.7	44.8	39.5	33.9	28.1	24.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.3	28.6	31.4	36.1	41.5	48.4	52.1	51.9	46.0	43.0	36.7	31.0		
		MCDB	31.2	31.3	33.5	40.1	45.5	52.9	56.9	56.6	50.7	48.0	40.6	32.8		
	2%	WB	25.1	26.2	28.9	33.1	39.2	46.4	49.0	49.2	43.9	40.1	33.1	29.0		
		MCDB	27.7	28.9	30.9	36.9	42.8	51.0	53.8	54.3	48.7	44.3	36.6	31.1		
	5%	WB	23.4	24.3	27.0	31.5	37.7	44.7	47.3	47.5	42.4	38.0	31.3	26.9		
		MCDB	26.3	27.0	28.8	34.0	40.9	48.6	52.0	52.4	46.4	41.2	34.2	28.9		
	10%	WB	21.7	22.3	25.0	30.2	36.3	43.1	45.9	45.9	41.0	36.0	29.8	24.6		
		MCDB	24.5	24.2	26.8	32.1	39.0	46.9	50.1	50.4	44.6	38.6	33.1	26.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.5	8.6	8.3	7.4	8.2	9.2	9.0	8.1	7.9	7.8	7.8		
		MCDBR	9.6	9.2	9.1	9.5	9.7	11.4	12.5	11.7	10.9	10.6	9.0	8.3		
		MCWBR	8.1	7.7	8.1	7.3	8.1	9.9	9.9	10.0	9.2	9.2	6.7	6.5		
	5% WB	MCDBR	9.1	8.8	8.7	8.4	8.8	10.5	11.0	10.8	9.6	9.5	7.4	7.3		
		MCWBR	8.5	8.3	8.3	7.3	7.9	10.0	9.8	10.2	8.6	9.1	6.8	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.222	0.239	0.268	0.305	0.304	0.302	0.302	0.307	0.291	0.265	0.242	0.225		
	taud		2.525	2.513	2.500	2.421	2.459	2.476	2.466	2.447	2.494	2.580	2.617	2.560		
	Ebn at Noon		306	315	314	306	307	307	306	302	301	300	296	296		
	Edh at Noon		21	26	30	35	35	34	34	34	30	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		533	792	1118	1434	1715	1936	2018	1826	1355	942	614	458		
	RadStd		78	102	117	165	128	165	146	150	109	111	60	69		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.10	N/A	N/A	N/A	+1.01	N/A	-409	-421	N/A	N/A
(47)	Regional (30 neighbors)	+0.85	N/A	N/A	N/A	+0.72	+0.72	N/A	-256	N/A	N/A

Nomenclature: See separate page